DC Motor Angular Speed Control hps5130

Eric Castro*, Esteban Segura* Daniel Chavarria*

*School of electronic engineering, Costa Rica Institute of Technology (ITCR), 30101 Cartago, Costa Rica,
{ecastro, essegura, dachavarria}@estudiantec.cr

***Abstract*—This paper presents the design and validation.....**
***Keywords*—DC Motor, controller, PI, stabilization, overshoot.**

I. Introduction

The control of angular velocity in motor of direct current, is a classic problem of automatic control with a big application in industrials systems and electric conditioning. A typical solution for this type of plant its the proportional-integrative regulator (PI), whose integrative action guarantee a zero steady-state error for inputs like step-type and the capacity of reject disturbances. [1]

In this experiment, it is designed a PI regulator for the CD motor system *hps5130*, modelling like a first order system, whose being identify via experimental. The design is performed for both continuous and discrete time with a sampling period of $T_s = 5\,\mathrm{ms}$, meeting the specifications of a settling time of less than 2% for 300 ms and overshoot of less than 3%. The results of both designs are verified by simulation in *Simulink*, including the response to disturbances apply to the plant's input and output. [3]

II. Theoretical Background

A. Motor CD as the control plant

The motor's DC current can be modeled as a first-order linear system in the frequency domain, whose transfer function is related to the armature voltage applied to the output angular speed:

$$G(s) = \frac{K}{s+p}, \tag{1}$$

where K is the static gain and p is the dominant pole, both obtained through experimental identification. [2]

B. PI regulator

The PI regulator can be expressed in continuous time as:

$$PI(s) = K_P + \frac{K_I}{s} = \frac{K_c(s - z_0)}{s}, \tag{2}$$

where K_P determines the system's response speed and K_I eliminates the steady-state error for step inputs. In its transfer function form, the PI introduces a pole at the origin $s = 0$ and a zero at $s = z_0$, whose relative location with respect to the plant pole directly influences the closed-loop system dynamics. [2]

C. Root-locus method

The root locus is a graphic technique that allows visualization of the displacement of the poles of the closed-loop in the complex plane s, in function of the controller gain. For a system with unity feedback, the poles of the closed-loop satisfy:

$$1 + K_c \cdot G_{OL}(s) = 0, \tag{3}$$

where $G_{OL}(s)$ is the open loop transfer function. The design of the regulator consist in locate the poles of closed-loop in the region's of the plane s that satisfy simultaneously the specifications of settling time and overshoot. [2] For a second-order system, these specifications impose restrictions on the real part and the angle of the dominant pole, expressed as:

$$\sigma = \frac{4}{t_s}, \tag{4}$$

$$\zeta = \frac{-\ln(M_p/100)}{\sqrt{\pi^2 + \ln^2(M_p/100)}}, \tag{5}$$

where σ is the minimum real part required for the dominant pole, t_s is the settling time to the 2% and ζ is the damping coefficient corresponding to the maximum overshoot M_p

D. Controller discretization

To implement a PI regulator in a system, the controller in the continuous time has to be convert into discrete time. Using the backward Euler approximation, the complex variable s is replaced by:

$$s \approx \frac{z-1}{T_s \cdot z}, \tag{6}$$

which transforms the continuous PI into its discrete equivalent:

$$PI(z) = \frac{b_1 z + b_0}{z + a_0}, \tag{7}$$

where the coefficients b_1, b_0 y a_0 depends of K_P, K_I and the sampling period T_s. The selection of T_s must guarantee between 30 and 75 samples during the stabilization time for an adequate representation of the system dynamics. [1] [2]

E. Performance Criteria

The design specifications are expressed in terms of three main metrics. The 2% settling time (t_s) is defined as the time required for the response to remain within a $\pm 2\%$ band around the final value. The maximum percent overshoot (M_p) represents the largest percentage deviation of the response with respect to that final value. Finally, the steady-state error is the difference between the reference value and the final output value under a step input, which is zero when the controller contains an integrator in its structure [1].

III. Methodology

This section outlines the design and experimental validation process of three PI controllers for the angular velocity regulation of the *hps5130* DC motor, in both continuous and discrete time.

A. Plant Model Selection

Multiple CSV datasets were collected from the *hps5130* motor during the identification experiment. A statistical analysis based on the mean and standard deviation of the recorded signals was performed to determine the most representative dataset, which was subsequently used to obtain the empirical first-order plant model via the *System Identification* toolbox in MATLAB, following the structure defined in (1).

B. PI Controller Design

Three PI controllers were designed in both continuous and discrete time using the *sisotool* root locus editor in MATLAB. Each design differed in the location of the real zero relative to the dominant plant pole, following the three placement criteria described in (2). All designs were constrained to meet a 2% settling time $t_s < 300\,\mathrm{ms}$ and a maximum percent overshoot $M_p < 3\,\%$. The continuous-time controllers were verified in *Simulink* under step input and disturbance rejection conditions, while the discrete-time equivalents were obtained at $T_s = 5\,\mathrm{ms}$ using the `c2d` function in MATLAB, but also, were verified in *Simulink* under step input and disturbance rejection conditions.

C. Experimental Validation

Falta esta parte, se tendría que completar luego del miércoles

IV. Results

This section presents the simulation and experimental results obtained for the three PI controllers designed in both continuous and discrete time.

A. Continuous-Time Simulation Results

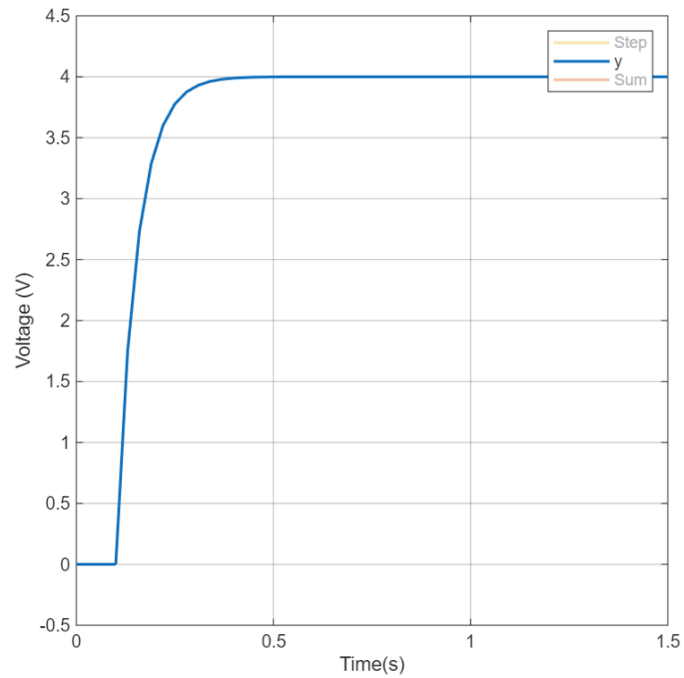

Figure 1. Step response of the three continuous-time PI controllers obtained in Simulink.

1) Step Response:

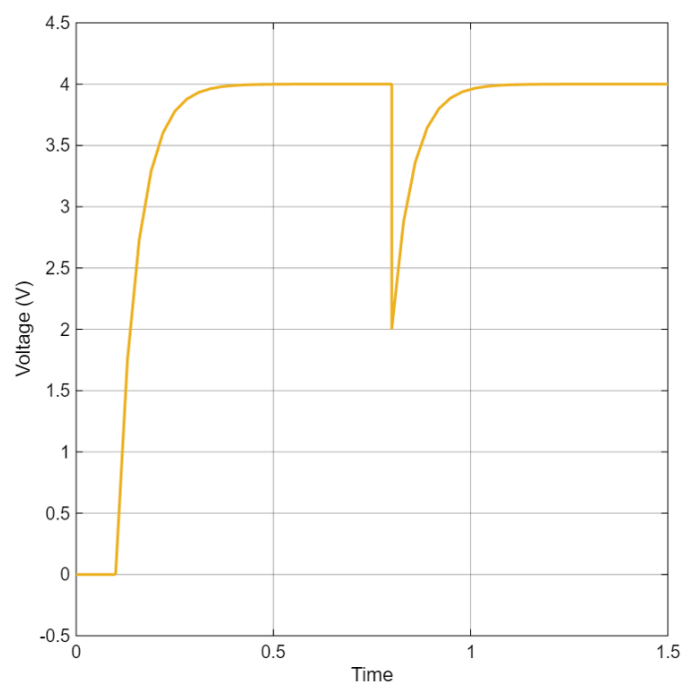

Figure 2. Disturbance rejection response of the three continuous-time PI controllers obtained in Simulink.

2) Disturbance Rejection:

B. Discrete-Time Simulation Results

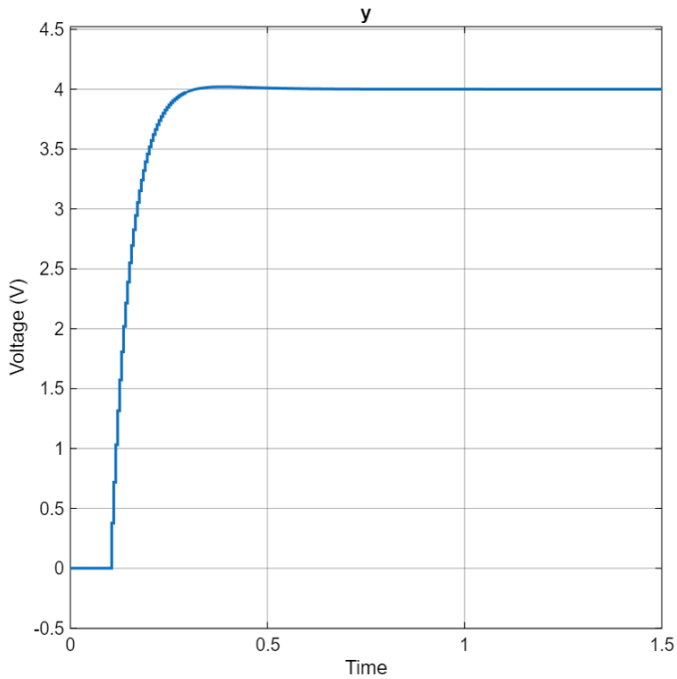

Figure 3. Step response of the best of three discrete-time PI controllers obtained in Simulink.

1) Step Response:

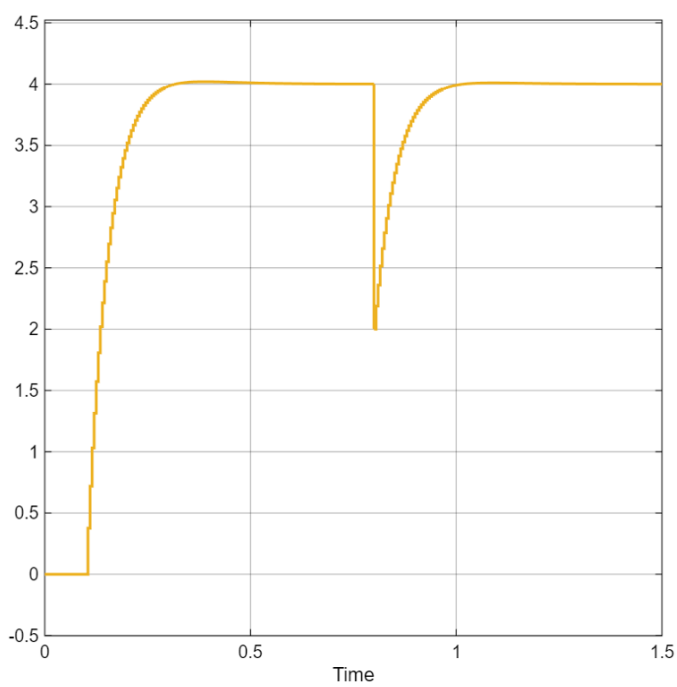

Figure 4. Disturbance rejection response of the best of three discrete-time PI controllers obtained in Simulink.

2) Disturbance Rejection:

C. Continuous-Time Experimental Results

figures/experimental_step.png

Figure 5. Experimental step response of the implemented PI controllers on the *kitprog059* platform on continuous-time.

1) Step Response:

figures/experimental_disturbance.png

Figure 6. Experimental disturbance rejection response of the implemented PI controllers on the *kitprog059* platform on continuous-time.

2) Disturbance Rejection:

D. Discrete-Time Experimental Results

figures/experimental_step.png

Figure 7. Experimental step response of the implemented PI controllers on the *kitprog059* platform on discrete-time.

figures/experimental_disturbance.png

Figure 8. Experimental disturbance rejection response of the implemented PI controllers on the *kitprog059* platform on discrete-time.

1) Disturbance Rejection:

E. Performance Summary

Table I
SIMULATION PERFORMANCE METRICS OF THE DESIGNED PI CONTROLLERS.

Controller	**Domain**	t_s **(ms)**	M_p **(%)**	e_{ss}
PI_1	Continuous	310	0	0
PI_2	Continuous			
PI_3	Continuous			
PI_1	Discrete	265	0.49	0
PI_2	Discrete			
PI_3	Discrete			

Table II
EXPERIMENTAL PERFORMANCE METRICS OF THE DESIGNED PI CONTROLLERS.

Controller	**Domain**	t_s **(ms)**	M_p **(%)**	e_{ss}
PI_1	Continuous			
PI_2	Continuous			
PI_3	Continuous			
PI_1	Discrete			
PI_2	Discrete			
PI_3	Discrete			

V. DISCUSSION

A. Statistical analysis

B. Experimental versus simulation results

VI. CONCLUSIONS

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